Linear Algebra Midterm Solutions

1. (a) Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}.$$

(b) Diagonalise A. (You can leave your eigenvalues as λ_1 and λ_2 in this.)

Solution

The zeros of the char poly satisfy

$$0 = (2-\lambda)(-\lambda) - 1 = \lambda^2 - 2\lambda - 1 = (\lambda - 1)^2 - 2$$

so are $\lambda_1 = 1 + \sqrt{2}$, and $\lambda_2 = 1 - \sqrt{2}$.

From the second row of $(A - \lambda_i I)$ we get that the ith eigenvector x_i, which must satisfy $(A - \lambda_i I)x_i = 0$, is $(\lambda_i, 1)$, so we have $x_1 = (\lambda_1, 1) = (1 - \sqrt{2}, 1)$, and $x_2 = (\lambda_2, 1) = (1 + \sqrt{2}, 1)$.

We can thus diagonalise A to

$$A = S\Lambda S^{-1} = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix}$$

2. Where the 2×2 matrix A has eigenvalues 1 and 3,

(a) what are the eigenvalues of $2A$?

(b) what are the eigenvalues of the 4×4 matrix

$$B = \begin{bmatrix} A & 4A \\ 0 & 2A \end{bmatrix}?$$

Solution

(a) The eigenvalues of $2A$ are 2 and 6: if λ is an eigenvalue of A with eigenvector x, then $2Ax = 2\lambda x$ so 2λ is an eigenvalue of $2A$.

(b) There is some invertible S that triangularise A: $T_A = S^{-1}AS$ for upper triangular T_A with the eigenvalues of A on the diagonal. So the invertible matrix

$$\begin{bmatrix} S & 0 \\ 0 & S \end{bmatrix}$$

triangularises B to

$$\begin{bmatrix} T_A & 4T_A \\ 0 & 2T_A \end{bmatrix}.$$

This has $1, 3, 2, 6$ along the diagonal, so these are the eigenvalues of B. (Can you find the eigenvectors of B? $(?, ?.x_1, x_2)$).

3. Let $G_{k+2} = 2G_{k+1} + G_k$.

   (a) Where A is the transition matrix such that for $x_k^T = [G_{k+1}, G_k]$, $x_{k+1} = Ax_k$ find an expression for A^n.

   (b) If $G_0 = 0$ and $G_1 = 2$, find a closed expression for G_n, (that is, solve the difference equation.)

Solution

A is the matrix from Question 1, so $A^n = S\Lambda^n S^{-1} =$

$$\begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{bmatrix} \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix}.$$

So

$$\begin{bmatrix} G_{n+1} \\ G_n \end{bmatrix} = x_n = A^n x_0 = S\Lambda^n S^{-1} \begin{bmatrix} 2 \\ 0 \end{bmatrix},$$

which expands to

$$\begin{bmatrix} G_{n+1} \\ G_n \end{bmatrix} = \frac{1}{2\sqrt{2}} \left(2\lambda_1^n \begin{bmatrix} \lambda_1^n \\ 1 \end{bmatrix} - 2\lambda_2^n \begin{bmatrix} \lambda_2^n \\ 1 \end{bmatrix} \right).$$

Thus $G_n = \frac{1}{\sqrt{2}}(\lambda_1^n - \lambda_2^n)$.

4. (a) What is the Hermitian A^H of the matrix

$$A = \begin{bmatrix} i & 4 \\ -1 & 2+i \end{bmatrix}?$$

   (b) What is a Hermitian matrix?

   (c) What is a Unitary matrix?

   (d) Show that if A can be diagonalised $A = U\Lambda U^{-1}$ by a unitary matrix U, then $AA^H = A^H A$.

Solution

$$A^H = \begin{bmatrix} -i & -1 \\ 4 & 2-i \end{bmatrix}?$$

A matrix A is Hermitian if $A = A^H$. It is Unitary if $AA^H = I = A^H A$. So for unitary matrixes U we have that $U^H = U^{-1}$. Thus if $A = U\Lambda U^{-1}$ for unitary U we have that

$$\begin{aligned} AA^H &= U\Lambda U^H (U\Lambda U^H)^H \\ &= U\Lambda U^H U\Lambda^H U^H \\ &= U\Lambda\Lambda^H U^H \\ &= U\Lambda^H \Lambda U^H \\ &= U\Lambda^H U^H U\Lambda U^H \\ &= A^H A. \end{aligned}$$

Where Λ and Λ^H commute because they are diagonal, (or are actually the same, as they are both the eigenvalue matrix of $A = A^H$.)

5. (a) Give the matrix M of the transformation taking the basis $\{e_1 = (1,0), e_2 = (0,1)\}$ to the basis $\{v_1 = \frac{1}{\sqrt{5}}(1,2), v_2 = \frac{1}{\sqrt{10}}(-4,2)\}$; that is, the matrix M such that $Me_1 = v_1$ and $Me_2 = v_2$.

Note

You will notice this problem has changed. Sorry, I meant to normalise the v_1 and v_2. Not doing so made this problem too difficult. (One student managed to solve it by deciding that I had made a mistake and meant M^T instead of M^{-1}, which would have been true if I had properly normalised the v_i. Well done to this student- you got a bonus mark for this.) With normalise v_i we have $M^T = M^{-1}$ so this is in the form we want it and we also have that M only rotates vectors, so doesn't change the circle. The only stretching of axes is done by the diagonal matrix in the middle.
I made the problem out of 1 instead of 3 and gave the point if you said anything.

I am changing it here so that it is more useful to study from.

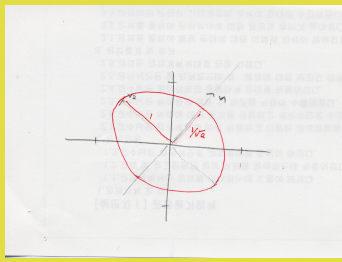
(b) Sketch the ellipse made up of the solutions

$$x^T M \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} M^{-1} x = 1.$$

Solution

The matrix M from the test was $M = \begin{bmatrix} 1 & -4 \\ 2 & 2 \end{bmatrix}$. For the changed version of the problem here, it is $M = \begin{bmatrix} 1/\sqrt{5} & -4/\sqrt{10} \\ 2/\sqrt{5} & 2/\sqrt{10} \end{bmatrix}$.

The matrix now multiplies the coefficeint of the v_1 term by 2, stretching the v_1 axis of the circle by $\frac{1}{\sqrt{2}}$. So the ellipse looks like



6. Prove that AB has the same eigenvalues as BA.

Solution

Let $\lambda \neq 0$ be an eigenvalue of AB with eigenvector x. Then $(BA)Bx = B(AB)x = B\lambda x = \lambda Bx$. So λ is an eigenvalue of BA with eigenvector Bx, (or Bx is 0 in which case $ABx = 0x$ contradicting the assumption that $\lambda \neq 0$).

If AB has an eigenvalue 0, then $ABx = 0$, so either A or B is singular. In which case BA also has eigenvalue 0. Thus AB and BA have the same eigenvalues.

Sorry. This last bit only works for square matrices. The statement is not true in the case that A and B are not square. If you had this, bring your test in and I'll give you some marks back.

7. (a) Find a 3×3 matrix A such that

$$[x_1, x_2, x_3]A[x_1, x_2, x_3]^T = 2(x_1 + x_3)(x_2 - 2x_3).$$

(b) Find another.

Solution

Expanding the polynomial, we get $2x_1x_2 - 4x_1x_3 + 2x_2x_3 - 4x_3^2$. So

$$\begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 1 \\ -2 & 1 & -4 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 2 & -4 \\ 0 & 0 & 2 \\ 0 & 0 & -4 \end{bmatrix}$$

are two such matrices.

8. (a) Give a necessary and sufficient condition for a matrix to be positive definite.

   (b) Show whether or not

$$\begin{bmatrix} 1 & 2 & 6 \\ 2 & 5 & 4 \\ 3 & 4 & 9 \end{bmatrix}$$

   is positive definite.

   (c) Show that if A is positive definite, then A^{-1} is.

Solution

A matrix A is positive definite it $x^T Ax > 0$ for all non-zero real vectors x
iff all pivots are positive
iff all eigenvalues are positive
iff all upperleft submatrices have positive determinants.

The determinant of the given matrix is $(45-16)-2(18-12)+6(8-15) = -1$ so no, it is not positive definite.

The eigenvalues of A^{-1} are $1/\lambda_i$ where λ_i are the eigenvalues of A (because A^{-1} shrinks the eigenvectors by the same factor that A stretches them), so are positive if the λ_i are. Thus A^{-1} is positive definite if A is.